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RTCA/DO-160G

Environmental Conditions and Test Procedures for Airborne Equipment

Section 11

Fluids Susceptibility

Important Notice

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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11.0 Fluids Susceptibility

11.1 Purpose of the Test

These tests determine whether the materials used in the construction of the equipment can withstand the deleterious effects of fluid contaminants. Fluids susceptibility tests should only be performed when the equipment will be installed in areas where fluid contamination could be commonly encountered. The fluids are representative of those commonly used fluids encountered in airborne and ground operations. Fluids not listed herein and for which susceptibility tests are indicated shall be included in the relevant equipment specification.

11.2 Precautions

Since many contaminants may have flash points within the test temperature range, care should be taken to ensure that adequate safety measures are taken to limit the possibility of fire or explosion.

Some contaminants may themselves or in combination with other contaminants or with the test sample be toxic. Due consideration should be given to this possibility before commencing the tests.

11.3 Equipment Categories

Category F

Equipment that has passed the tests covered in this section is identified as Category F. Details of the test fluids involved and the methods used shall be provided in the Environmental Qualification Form (See Appendix A).

Note: Sections 10.0 and 14.0 of this document cover waterproofness and salt spray tests, respectively. Section 11.0 covers seven general classes of other contaminating fluids. In addition there are 25 specific fluids that are used to test these classes. Table 11-1 contains the class of fluids, the specific fluids and the temperatures required in these tests. Testing of all 25 fluids or all 7 general classes of contaminated fluids is not required to claim category F. It is required to specify fluids used, test method and temperature of exposure in the test procedure and the Environmental Qualification Form.

11.4 Test Procedures

11.4.1 Spray Test

Connect the equipment mechanically and electrically as defined in the relevant equipment specification. The equipment is not required to operate during this test and shall be at room ambient.

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(a) Spray the specified fluid(s) to the entire surface of the test item that is likely to be exposed. The spray shall be a fine mist maintained at the temperatures in Table 11-1 and shall be directed toward every major surface, seal and connector of the equipment sample under test. Maintain surfaces in a wetted condition for 8 hours followed by a drying period of 16 hours at 65°C. If it is difficult to maintain a wetted condition and the equipment specification requires the spray test rather than the immersion test, it shall be acceptable to thoroughly spray the equipment at intervals of four hours maximum, and allow it to stabilize at standard ambient temperature.

(b) Visually examine the test item for degradation of materials, protective finishes, and physical changes. Repeat Step (a) until three 24-four hours cycles have been completed. At the end of the third spray period, while still wet, the product shall be operated for 10 minutes minimum. After the last cycle, ramp the chamber to standard ambient temperature (avoiding temperature shock) and operate the unit until stabilized or for a minimum of 30 minutes. DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

(c) Repeat steps (a) and (b) for additional fluids.

(d) After the completion of testing all of the fluids. DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

Note: If the equipment is to be tested with more than one class of contaminating fluid, it should normally be tested with each fluid separately. However, simultaneous testing is permitted if fluids are of the same base (Example: oil base should not be followed by water base fluid). Fluids should not be pre-mixed prior to spraying, and the order of application should be as specified in the equipment specification. Unless otherwise noted in the equipment specification, the total exposure time for simultaneous application of fluids should be the same as the exposure time for a single fluid. The precautions noted elsewhere in this section should be observed.

At the completion of the test, the equipment shall be inspected and shall show no evidence of damage of any internal or external component.

11.4.2 Immersion Test

Connect the equipment mechanically and electrically as defined in the relevant equipment specification. The equipment is not required to operate during this test and shall be at ambient temperature. Do not pre-mix any solution.

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Immerse the equipment in the appropriate fluid for a minimum of 24 hours. The fluid temperature shall be maintained at the temperature shown in Table 11-1 and shall cover the test specimen completely.

At the end of 24 hours, operate the equipment for at least 10 minutes while it is completely immersed in the fluid.

Following this period, remove the test specimen, place in an appropriate chamber and subject it to a constant temperature of +65 degrees C for minimum of 160 hours. At the end of this period, the test specimen shall be returned to room temperature and operated for a minimum of two hours.

Following the two-hour period, DETERMINE COMPLIANCE WITH THE APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.
At the completion of the test, the equipment shall be inspected and shall show no evidence of damage of any internal or external component.

11.5 Use of Material Specimens

Material specimen test may be used in place of equipment tests. The results of these tests shall assure that the material will protect the equipment from deleterious effects after being exposed to the relevant fluid in the manner defined in the equipment test procedures (Subsection 11.4).

Note: Material specimen tests are not permitted if the equipment is to be subsequently subjected to the continuous stream proof test (paragraph 10.3.3).

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Table 11-1 Classes of Test Fluids and Fluid Temperatures

Class of Contaminating Fluid	Test Fluid	Fluid Temperature Degrees C
Fuels	Aviation Jet A Fuel Aviation Piston Engine Fuel	40 1/ 40 1/
Hydraulic Fluids	Mineral-Based Non-mineral Based Phosphate Ester-Based (Synthetic), Type IV 2/ Silicate Ester-Based (Synthetic) Silicone-Based (Synthetic) Synthetic Hydrocarbon Base	80 50 70 70 70 70
Lubricating Oils	Mineral-Based Ester-Based (Synthetic) Internal Combustion Engine 15W40	70 150 70
Solvents and Cleaning Fluids	Isopropyl Alcohol Denatured Alcohol Cleaning compound for aircraft surfaces Trans-1-2Dichloroethylene (replacement for 1,1,1 Trichloroethylene)	50 1/ 23 1/ 23 23
De-Icing Fluid	Ethylene Glycol Propylene Glycol AEA Type 1 3/ AEA Type 2 3/ SAE Type 1 SAE Type 2 SAE Type 4	50 50 50 50 23 23 23
Insecticides	Dichlorvos (DDVP) (commercially available min 0.92% concentration by volume) Pyrethroid Pesticide (commercially available min 0.92 concentration by volume)	23 23
Sullage	To be defined by the equipment specification	23
Disinfectant (Heavy duty Phenolics)	Clear, soluble phenolics, e.g., phenol or its derivatives dissolved in surfactant and diluted with water to give a clear solution. Black fluids, e.g., refined tar products dissolved in a carrier oil and emulsified with detergent. White fluids, e.g., colloidal emulsions of refined coal tar products in water, usually containing a small amount of surfactant.	23 23 23
Coolant dielectric fluid	PAO dielectric	70

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Fire extinguishants	Protein Fluoroprotein Aqueous Film Forming Foam (ATTF)	23 23 23

NOTES – For Table 11-1:

1/ This temperature exceeds the critical flash point temperature. Testing should always be performed in a suitable pressure vessel.

2/ These fluids are electrically conductive. Suitable precautions should be taken after exposure to the fluids before operating the equipment.

3/ Association of European Airlines.

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